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9 February 2025  
QTW- Module 6 Naive Bayes – Presssion

Relevant Questions:

1. Why does Naive Bayes often perform well in high-dimensional text classification, despite the naive independence assumption?
2. How does the independence assumption in Naive Bayes affect its performance on real-world datasets where features are often correlated?
3. In what scenarios would we choose Naive Bayes over other classification algorithms?

Takeaways:

1. The Independence Assumption's Paradox
  - Despite its "naive" assumption that features are independent (which is rarely true in real-world data), Naive Bayes often performs surprisingly well in practice
  - This is particularly true in text classification and high-dimensional problems
  - Understanding: The model's simplicity and efficiency often outweigh the limitations of the independence assumption
  - Mathematical representation:  $P(x_1, x_2 | y) = P(x_1 | y) \times P(x_2 | y)$
2. Probabilistic Foundation and Scalability
  - Naive Bayes is fundamentally a probabilistic classifier based on Bayes' Theorem
  - It's highly scalable and can handle large datasets efficiently
  - Training is  $O(n)$  complexity, making it one of the fastest classifiers
  - Key equation:  $P(y|x) = P(x|y)P(y)/P(x)$
  - This makes it particularly valuable for real-time applications and large-scale classification tasks
3. Susceptibility to Zero Probability Problem
  - When a feature value in the test data never appears in the training data, it leads to zero probability
  - This is solved through smoothing techniques (like Laplace/Additive smoothing)
  - Critical for practical implementation:  $P(x|y) = (\text{count}(x,y) + \alpha) / (\text{count}(y) + \alpha|V|)$
  - Understanding this limitation and its solutions is crucial for effective implementation